

YASHPAL SINGH CHOUHAN

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EDUCATION

Vellore Institute of Technology (VIT), Bhopal

2022 – 2027 (Expected)

Integrated M.Tech in Artificial Intelligence

CGPA: 8.10

Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, NLP, DBMS, OOP

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript (ES6+), SQL
- **AI / LLM:** LangChain, Agentic AI, RAG Pipelines, Prompt Engineering, Gemini API, Groq/LLaMA 3, HuggingFace Transformers, Sentence Transformers, ChromaDB, FAISS
- **Backend & Frontend:** FastAPI, REST API, React, Next.js, Tailwind CSS, Streamlit
- **Data & Tools:** Pandas, NumPy, Matplotlib, spaCy, Git/GitHub, Docker, Vercel, Render

EXPERIENCE

CortexGiga, Remote

Jan 2026 – Mar 2026

AI/ML Intern

- Developed **NLP pipelines** using **FastAPI** and **HuggingFace**, reducing manual data processing time by **40%** through automated **BERT-based** text classification across **3 production workflows**.
- Fine-tuned **transformer-based models** for text classification and generation tasks, improving inference efficiency by **35%** through quantization and batching strategies.
- Architected **4 REST APIs** using **FastAPI** and **Pydantic** with asynchronous processing, sustaining **100+ requests per second** for real-time inference under concurrent load.

PROJECTS

Medical RAG Assistant

GitHub

Python, FastAPI, LangChain, ChromaDB, Gemini API, HuggingFace, Sentence Transformers, PyMuPDF

- Built an end-to-end **RAG pipeline** parsing a **173-page clinical handbook** into **532 semantic vectors** (PyMuPDF + LangChain + ChromaDB + HuggingFace embeddings), enabling context-aware retrieval with sub-second latency.
- Exposed a **FastAPI endpoint** for ingestion, semantic search, and **Gemini-powered answer generation** with page-level citations — reducing hallucinations to zero on in-corpus queries.

Indian News Bias Detector

GitHub

Python, FAISS, LangChain, Groq/LLaMA 3.3 70B, spaCy, Sentence Transformers, Streamlit

- Built an **NLP pipeline** over **300+ articles** from 4 outlets across 15 topics — combining VADER sentiment, TextBlob subjectivity, and spaCy NER for outlet-level bias scoring.
- Paired a **FAISS vector store** (SentenceTransformer embeddings) with **Groq LLaMA 3.3 70B** to generate grounded bias explanations; visualized via a Streamlit dashboard with heatmaps and topic-wise comparisons.

DSA AI Coach

GitHub | Live

Next.js, React, TypeScript, Tailwind CSS, FastAPI, Gemini API, Vercel, Render

- Shipped a **full-stack AI tutoring platform**: Next.js frontend (Vercel) + FastAPI backend (Render); **Gemini API** generates **adaptive roadmaps** and lesson content from real-time learner performance data.
- Engineered **Socratic hint delivery** via prompt engineering — LLM infers the user's solution approach and surfaces targeted hints over direct answers, improving conceptual retention.

CERTIFICATIONS & ACHIEVEMENTS

- **Applied Machine Learning in Python** — University of Michigan (Coursera)
- **Computer Vision** — VIT Bhopal University (VITYarthi)
- Solved **100+ DSA problems** on LeetCode / Codeforces: arrays, graphs, dynamic programming, tree traversal
- Research Intern, Kanchan Children Welfare & Educational Society — conducted field assessments across **5+ rural schools**; findings shaped NGO program planning and resource allocation